

DUNHUANG, China

WMO: 524180

Lat: 40.150N

Lon: 94.683E

Elev: 1140

StdP: 88.36

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-15.8	-25.6	0.4	-8.9	-23.6	0.5	-8.9	7.9	-6.3	7.0	-6.5	0.8	180	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	35.5	18.2	34.2	17.7	32.9	17.0	20.3	31.8	19.4	30.6	18.6	29.5	2.3	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	13.8	23.9	15.8	12.9	23.9	14.7	12.0	24.1	64.2	32.0	60.6	30.6	57.7	29.7	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.6	6.4	5.4	DB	-20.8	38.3	2.1	1.4	-22.3	39.3	-23.5	40.1	-24.7	40.9	-26.2	41.9
			WB	-21.2	22.8	2.0	1.1	-22.6	23.6	-23.8	24.2	-25.0	24.8	-26.5	25.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-7.5	-1.6	6.3	13.8	19.4	24.0	25.9	24.2	18.5	10.6	2.3	-5.6		
	DBStd	12.02	3.42	4.22	4.46	4.34	3.55	2.63	2.46	2.53	3.10	3.49	4.02	3.73		
	HDD10.0	1764	542	324	129	15	0	0	0	0	1	37	231	485		
	HDD18.3	3407	801	558	373	144	30	1	0	1	34	241	481	743		
	CDD10.0	2102	0	0	14	130	293	419	493	442	257	54	0	0		
	CDD18.3	703	0	0	0	9	64	170	234	184	41	0	0	0		
	CDH23.3	8399	0	0	2	173	880	1997	2719	2061	540	27	0	0		
	CDH26.7	3809	0	0	0	41	315	934	1391	977	151	2	0	0		
(14) Wind	WSAvg	2.0	1.9	2.1	2.4	2.6	2.3	2.1	2.0	1.8	1.7	1.6	1.8	1.9		
(15) Precipitation	PrecAvg	38	1	1	1	2	3	8	12	6	2	1	1	0		
	PrecMax	105	6	4	11	16	13	35	61	29	12	5	12	3		
	PrecMin	9	0	0	0	0	0	0	1	0	0	0	0	0		
	PrecStd	21	1	1	3	4	4	9	13	7	3	1	2	1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	13.1	22.8	30.7	33.6	35.9	38.0	36.6	32.4	26.3	17.1	6.6		
		MCWB	-1.0	3.2	8.3	12.4	14.1	17.0	19.2	19.2	15.8	11.4	6.3	0.1		
	2%	DB	2.6	10.1	20.0	27.4	31.8	34.4	36.0	34.7	30.1	23.5	14.4	4.2		
		MCWB	-2.7	1.6	6.9	10.8	13.8	16.7	18.8	18.1	14.7	10.5	4.8	-1.4		
	5%	DB	0.5	8.1	17.6	25.2	30.1	33.1	34.6	33.2	28.4	21.3	12.3	2.4		
		MCWB	-3.9	0.6	5.8	9.8	13.0	16.2	18.6	17.6	14.1	9.4	3.9	-2.4		
	10%	DB	-1.2	6.1	15.3	23.1	28.3	31.8	33.2	31.8	26.6	19.2	10.0	0.7		
		MCWB	-5.0	-0.5	4.8	8.9	12.3	15.9	18.1	17.3	13.5	8.3	2.8	-3.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.9	3.5	8.7	13.0	16.3	20.4	22.4	21.2	17.7	12.5	6.6	0.3		
		MCDB	4.9	12.2	22.0	28.7	30.3	31.3	34.1	32.8	27.7	22.8	16.2	6.0		
	2%	WB	-2.5	1.9	7.1	11.3	14.6	18.9	20.8	19.9	16.2	11.1	5.1	-1.2		
		MCDB	2.2	9.5	19.1	26.1	28.8	30.1	32.5	30.8	26.6	22.2	13.7	3.7		
	5%	WB	-3.8	0.7	6.0	10.2	13.6	17.9	19.8	19.0	15.2	10.0	4.0	-2.3		
		MCDB	0.3	7.7	17.1	23.9	27.7	29.4	31.3	29.7	25.4	20.3	12.0	2.0		
	10%	WB	-4.9	-0.4	4.9	9.2	12.7	17.0	19.0	18.2	14.3	8.8	2.9	-3.4		
		MCDB	-1.3	5.8	15.0	21.9	26.6	28.1	30.4	28.9	24.5	18.2	9.8	0.5		
(35) Mean Daily Temperature Range	MDBR		12.1	13.7	14.5	14.9	14.5	13.8	13.1	14.2	15.8	16.7	13.3	11.3		
	5% DB	MCDBR	13.9	15.7	17.8	17.7	17.6	16.4	15.8	16.8	18.0	19.1	16.4	12.9		
		MCWBR	9.9	9.6	8.8	7.2	6.7	6.3	6.5	6.6	6.8	8.2	8.9	8.6		
	5% WB	MCDBR	13.4	15.1	17.2	16.3	15.4	14.0	13.9	14.8	15.4	18.2	16.0	12.4		
		MCWBR	9.6	9.3	8.6	6.7	6.5	6.4	6.5	6.4	6.4	7.8	8.8	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.324	0.381	0.442	0.501	0.499	0.517	0.508	0.475	0.448	0.395	0.354	0.326		
	taud		2.133	1.988	1.843	1.741	1.797	1.823	1.893	1.958	1.965	2.016	2.080	2.125		
	Ebn at Noon		811	806	795	777	790	775	778	792	782	779	761	770		
	Edh at Noon		107	144	187	221	214	209	193	176	163	138	112	100		
(44) All-Sky Solar Radiation	RadAvg		2.67	3.70	5.11	6.27	7.10	7.14	6.87	6.43	5.53	4.38	3.03	2.37		
	RadStd		0.19	0.15	0.22	0.32	0.32	0.43	0.27	0.32	0.26	0.14	0.13	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.38	N/A	N/A	N/A	-0.44	N/A	N/A	-84	+110	+70
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-106	+84	+31

Nomenclature: See separate page